

State Machine Diagram – AR Accessibility Assistant

This document explains the State Machine Diagram for the DetectionSession object in an AR-based accessibility system.

Key States:

- 1 Idle – System waiting for user input
- 2 Initializing Camera – Camera setup phase
- 3 Scanning – Capturing real-time frames
- 4 Processing Image – AI model analyzing frame
- 5 Object Detected – Object successfully recognized
- 6 No Object Found – No valid object detected
- 7 Generating Audio – Converting result to speech
- 8 Completed – Output delivered to user
- 9 Error – System failure state

Transitions:

- 1 User action starts the detection process
- 2 Camera initialization leads to scanning
- 3 Captured frame moves system to processing
- 4 AI confidence determines detection success
- 5 Detected object triggers audio generation
- 6 Audio output leads to completion
- 7 Any failure leads to error state

Importance:

This diagram helps visualize system behavior, improve debugging, and ensure smooth interaction for visually impaired users.